ELECTRONICS LAB / ECED / VERSION 1.1

Worked examples and worksheets

Typeset equations, circuit diagrams and student observation sheets.

15 live lessons | 4 November 2026

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Use the interactive page to change values and test predictions.

These pages capture representative examples; observation spaces are left blank.

AC quantities use RMS unless explicitly labelled peak or instantaneous.

Ideal models support laboratory preparation and comparison with measurements.

ECED / Module 1 · DC circuit fundamentals / LIVE LESSON

Ohm’s law and dividers

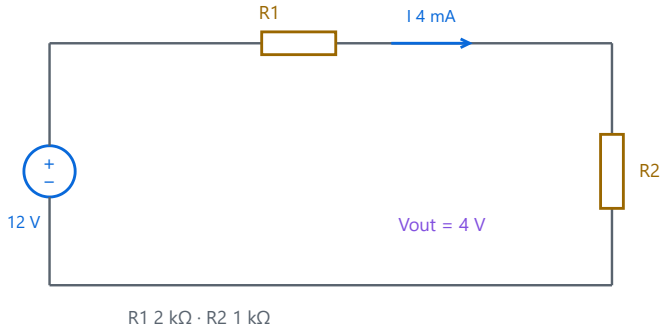
Compare series and parallel resistor networks. Explore voltage division, current division, V–I relationships and power balance.

CO1 · Lab 1 preparation · Prerequisite: none · Suggested activity: 15 minutes · Version 1.1 · Typeset mathematics

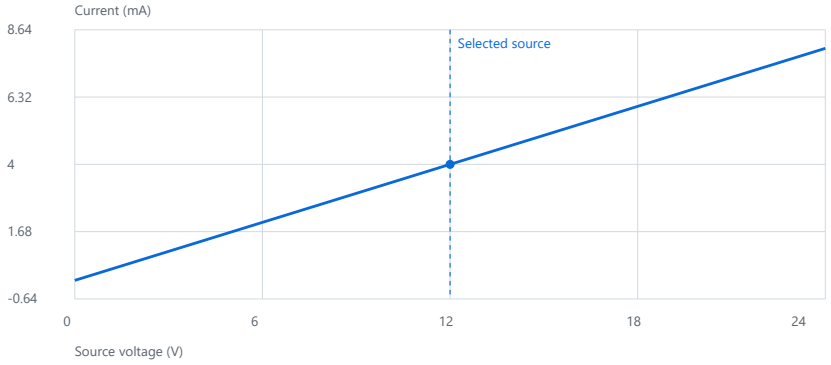
Equivalent resistance	Source current	Total resistor power
3 kΩ	**4 mA**	**48 mW**
series	I = V / Req	P = V × I

Same current, divided voltage

Change the connection and compare the circuit equations. The graph sweeps source voltage while holding both resistances fixed.



R1 2 kΩ · R2 1 kΩ



● Total current

Series: both resistors carry the same current. Vout across R2 depends on the ratio R2/(R1+R2). An added load would change that ratio.

Calculation step by step

RMS values for AC; SI units internally. Display rounding does not affect the calculations.

1 / CONNECTION RULE

$$I_1 = I_2 = I_s$$

2 / EQUIVALENT RESISTANCE

$$R_{eq} = R_1 + R_2 = 3\,\text{k}\Omega$$

3 / OHM'S LAW

$$I_s = \frac{12}{3{,}000} = 4\,\text{mA}$$

$$I_1 = 4\,\text{mA}; \quad I_2 = 4\,\text{mA}$$

4 / DIVIDER RELATION

$$V_{out} = \frac{V_s R_2}{R_1 + R_2} = 4\,\text{V}$$

$$V_1 = \frac{V_s R_1}{R_1 + R_2} = 8\,\text{V}$$

5 / CIRCUIT-LAW CHECK

$$V_1 + V_2 - V_s = 0\,\text{V}$$

6 / POWER CHECK

$$P_1 = I_1^2 R_1 = 32\,\text{mW}$$

$$P_2 = I_2^2 R_2 = 16\,\text{mW}$$

$$P_1 + P_2 = V_s I_s = 48\,\text{mW}$$

Check your understanding

Commit to a prediction, then check the explanation.

1. In a series circuit, which quantity is the same in both resistors?

- ◯ Current
- ◯ Voltage

---

2. For a fixed voltage source and parallel resistors, doubling R1 changes...

- ◯ Both branch currents by the same amount
- ◯ I1 and total current; I2 stays fixed

3. Numeric practice

A 12 V source feeds series R1=2 kΩ and R2=1 kΩ. Find the unloaded voltage across R2 in volts.

Your answer

Observation worksheet

Compare the series and parallel presets. Record Req, total current, branch currents and voltage across R2. Double R1 and explain the change. Supply setting: 12 V; R1=2000 Ω; R2=1000 Ω.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> Ideal DC source and two positive resistors. Series output is measured across R2 with no extra load attached. Parallel branches both receive the full source voltage. Current-divider ratios refer to the total current entering the two branches; changing a resistor with a fixed voltage source also changes the total current. Rectangular IEC resistor symbols are used. Supports preparation for Lab 1.

ECED / Module 1 · DC circuit fundamentals / LIVE LESSON

KCL and KVL

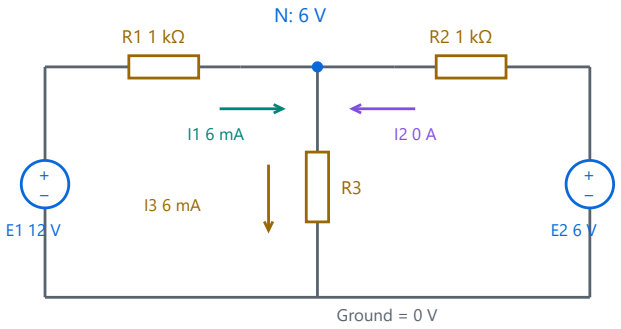
Solve a two-source resistor network. Inspect signed currents, node voltage, loop sums and power balance.

CO1 · Lab 1 · Prerequisite: Ohm's law · Suggested activity: 15 minutes

Node voltage	Load current I3	Resistor power
6 V	**6 mA**	**72 mW**
relative to ground	I1 + I2 = I3	sum of three resistors

Watch the node and loop balances

Arrows define positive reference directions. Negative current means flow opposite the arrow. Resistors use IEC rectangular symbols.



The node balances. The signed currents entering N equal the current leaving through R3. Both loop sums are zero within numerical precision.

Calculation step by step

RMS values for AC; SI units internally. Display rounding does not affect the calculations.

1 / WRITE KCL

$$\frac{E_1 - V_N}{R_1} + \frac{E_2 - V_N}{R_2} = \frac{V_N}{R_3}$$

2 / SOLVE THE NODE VOLTAGE

$$V_N = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

$$V_N = 6\,\mathrm{V}$$

3 / CALCULATE BRANCH CURRENTS

$$I_1 = \frac{12 - 6}{1{,}000} = 6\,\mathrm{mA}$$

$$I_2 = \frac{6 - 6}{1{,}000} = 0\,\mathrm{A}$$

$$I_3 = \frac{V_N}{R_3} = 6\,\mathrm{mA}$$

4 / WALK EACH LOOP

Left: $E_1 - I_1 R_1 - I_3 R_3 = 0\,\mathrm{V}$

Right: $E_2 - I_2 R_2 - I_3 R_3 = 0\,\mathrm{V}$

5 / CHECK KCL

$$I_1 + I_2 - I_3 = 0\,\mathrm{A}$$

Keep the signs. Do not add only current magnitudes.

6 / CHECK POWER

$$\sum I^2 R = 72\,\mathrm{mW}$$

$$E_1 I_1 + E_2 I_2 = 72\,\mathrm{mW}$$

Positive source power here means delivered power. A negative contribution means absorption.

Check your understanding

Commit to a prediction, then check the explanation.

1. A computed I2 is negative. What does that mean?

- ◯ KCL has failed
- ◯ I2 flows opposite its chosen arrow
- ◯ R2 must have negative resistance

2. Can an ideal voltage source absorb power in this network?

- ◯ Yes, depending on the signed current
- ◯ No, a source can only deliver power

3. A worked numeric check

For E1=12 V, E2=6 V and all three resistors 1 kΩ, find VN in volts.

Your answer

Observation worksheet

Set E1=12 V, E2=6 V and R1/R2/R3=1000/1000/1000 Ω. Record VN, I1, I2, I3 and the two loop sums. Change E2 to 0 V and predict I2 before checking.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> Ideal DC voltage sources and positive resistors. Ground is the common lower rail. I1 and I2 point into node N; I3 points toward ground. A negative branch current flows opposite its labelled reference. Supports Lab 1 (KCL/KVL using LTspice); compare signed branch currents and loop drops.

ECED / Module 1 · DC circuit fundamentals / LIVE LESSON

Node analysis

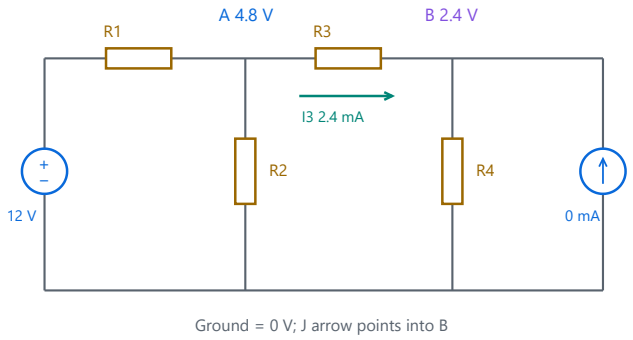
Write and solve a two-node conductance matrix. Inspect signed branch currents and verify both KCL equations.

CO1 · Lab 2 · Prerequisite: KCL and Ohm's law · Suggested activity: 20 minutes · Version 1.1 · Typeset mathematics

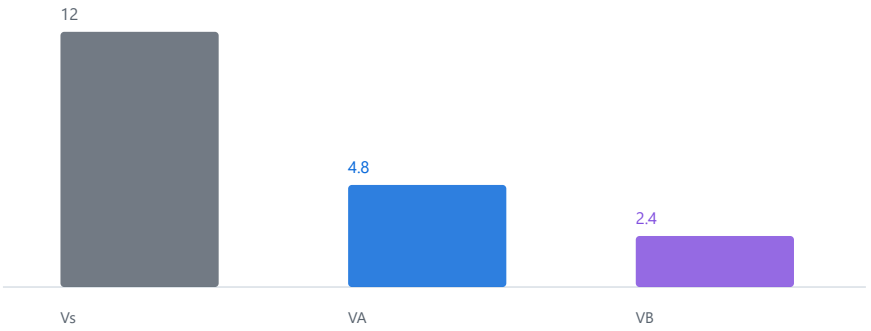
Node A voltage	Node B voltage	Bridge current I3
4.8 V	**2.4 V**	**2.4 mA**
VA relative to ground	VB relative to ground	positive from A to B

One reference node, two unknown voltages

I1 enters A from the source; I2 leaves A to ground; I3 goes from A to B; I4 leaves B to ground. The current-source arrow defines positive injection.



Ground = 0 V; J arrow points into B



Node equations balance simultaneously. Solving VA and VB together determines all branch currents; they are not independent voltage dividers.

Calculation step by step

RMS values for AC; SI units internally. Display rounding does not affect the calculations.

1 / WRITE KCL AT A

$$\frac{V_\mathrm{A} - V_\mathrm{s}}{R_1} + \frac{V_\mathrm{A}}{R_2} + \frac{V_\mathrm{A} - V_\mathrm{B}}{R_3} = 0$$

2 / WRITE KCL AT B

$$\frac{V_\mathrm{B} - V_\mathrm{A}}{R_3} + \frac{V_\mathrm{B}}{R_4} = J$$

J is in amperes in the equation. Positive J enters B from the current source.

3 / FORM THE MATRIX

$$\begin{bmatrix} g_1 + g_2 + g_3 & -g_3 \\ -g_3 & g_3 + g_4 \end{bmatrix} \begin{bmatrix} V_A \\ V_B \end{bmatrix} = \begin{bmatrix} V_s g_1 \\ J \end{bmatrix}$$

gk=1/Rk; values below in mS and mA:

$$\begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} V_A \\ V_B \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \end{bmatrix}$$

4 / SOLVE

$$a = g_1 + g_2 + g_3; \quad b = g_3 + g_4$$

$$b_A = V_s g_1; \quad b_B = J$$

$$\det = ab - g_3^2 = 0.000005\,\mathrm{S}^2$$

$$V_\mathrm{A} = \frac{b_A b + g_3 b_B}{\det} = 4.8\,\mathrm{V}$$

$$V_\mathrm{B} = \frac{a b_B + g_3 b_A}{\det} = 2.4\,\mathrm{V}$$

5 / RECOVER CURRENTS

$$I_1 = \frac{V_\mathrm{s} - V_\mathrm{A}}{R_1} = 7.2\,\mathrm{mA}$$

$$I_2 = \frac{V_\mathrm{A}}{R_2} = 4.8\,\mathrm{mA}$$

$$I_3 = \frac{V_\mathrm{A} - V_\mathrm{B}}{R_3} = 2.4\,\mathrm{mA}$$

$$I_4 = \frac{V_\mathrm{B}}{R_4} = 2.4\,\mathrm{mA}$$

6 / VERIFY THE SOLUTION

A: $I_1 - I_2 - I_3 = 0\,\mathrm{A}$

B: $I_3 + J - I_4 = 0\,\mathrm{A}$

Σ I^2R = 86.4 mW; sources deliver 86.4 mW

Check your understanding

Commit to a prediction, then check the explanation.

1. How many unknown node voltages does this grounded-source circuit need?

- ◯ Two: VA and VB
- ◯ Three, including the known source node

2. VA is below VB. What is the sign of I3 defined from A to B?

- ◯ Positive
- ◯ Negative

3. Numeric practice

For Vs=12 V, J=0 and all four resistors 1 kΩ, find VA in volts.

Your answer

Observation worksheet

Write both KCL equations, then compare them with the matrix solution. Record VA, VB and all four currents. Use the injection preset and explain bridge-current reversal. Present values: Vs=12 V; J=0 mA.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> Two unknown node voltages VA and VB, one grounded ideal voltage source, four positive resistors, and an independent current source from ground into B. J may be negative, meaning actual current is opposite its reference arrow. A voltage-source node connected to ground is known and is not an extra unknown. This fixed topology does not require a supernode. Supports node-analysis preparation for Lab 2.

ECED / Module 1 · DC circuit fundamentals / LIVE LESSON

Mesh analysis

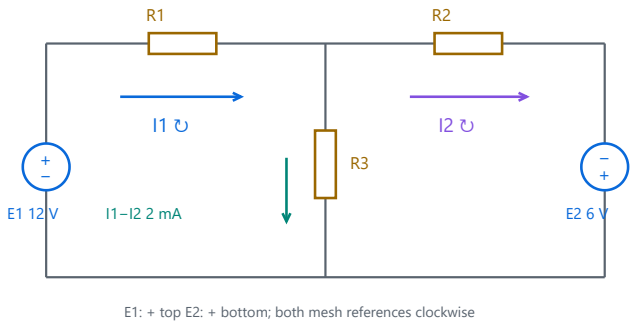
Solve a two-mesh network, track source polarities, and distinguish mesh currents from shared-branch current.

CO1 · Lab 2 · Prerequisite: KVL and simultaneous equations · Suggested activity: 20 minutes · Version 1.1 · Typeset mathematics

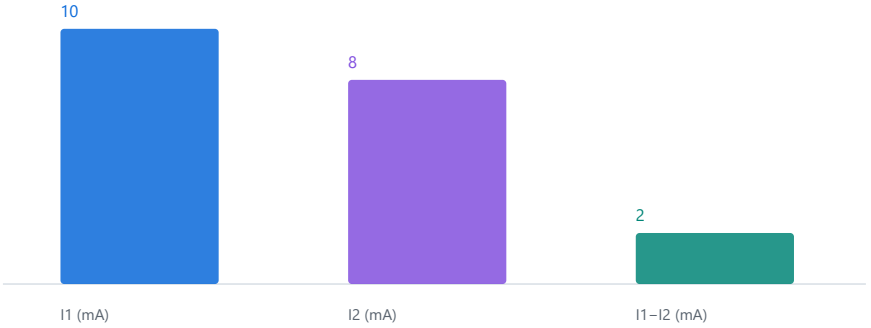
Left mesh I1	Right mesh I2	Shared R3 current
10 mA	**8 mA**	**2 mA**
clockwise reference	clockwise reference	downward reference: I1 − I2

Mesh currents are not all branch currents

The two clockwise mesh references pass through R3 in opposite directions. Read the source polarities before writing the KVL equations.



E1: + top E2: + bottom; both mesh references clockwise



R3 carries the mesh-current difference. I1 is larger, so the actual current through R3 is downward.

Calculation step by step

RMS values for AC; SI units internally. Display rounding does not affect the calculations.

1 / LEFT-LOOP KVL

$$-E_1 + R_1 I_1 + R_3 (I_1 - I_2) = 0$$

$$(R_1 + R_3) I_1 - R_3 I_2 = E_1$$

2 / RIGHT-LOOP KVL

$$-E_2 + R_2 I_2 + R_3 (I_2 - I_1) = 0$$

$$-R_3 I_1 + (R_2 + R_3) I_2 = E_2$$

E2 is a rise in the clockwise walk because its + terminal is at the bottom.

3 / MATRIX AND SOLUTION

$$\begin{aligned} a &= R_1 + R_3; \quad b \\ &= R_2 + R_3; \quad D \\ &= ab - R_3^2 \end{aligned}$$

$$\begin{bmatrix} a & -R_3 \\ -R_3 & b \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} E_1 \\ E_2 \end{bmatrix}$$

$$I_1 = \frac{E_1 b + E_2 R_3}{D} = 10\,\text{mA}$$

$$I_2 = \frac{E_2 a + E_1 R_3}{D} = 8\,\text{mA}$$

4 / BRANCH CURRENT

$$\begin{aligned} I_{R1} &= I_1; \quad I_{R2} \\ &= I_2; \quad I_{R3} \\ &= I_1 - I_2 \end{aligned}$$

$$I_{R3} = 2\,\text{mA}$$

$$\text{Top-node voltage} \quad = I_{R3} R_3 = 2\,\text{V}$$

5 / KVL RESIDUALS

$$\text{Left loop} \quad = 0\,\text{V}$$

$$\text{Right loop} \quad = 0\,\text{V}$$

6 / POWER BALANCE

$$P_1 + P_2 + P_3 = 168\,\text{mW}$$

$$E_1 I_1 + E_2 I_2 = 168\,\text{mW}$$

Source terms use the stated voltage-rise and current references.

Check your understanding

Commit to a prediction, then check the explanation.

1. What is the downward current through the shared resistor?

- ◯ I1+I2
- ◯ I1−I2

2. If I1=I2≠0, what is the shared resistor's current?

- ◯ Zero
- ◯ Equal to either mesh current

3. Numeric practice

E1=12 V, E2=6 V, R1=R2=R3=1 kΩ, with the polarities shown. Find downward shared current in mA.

Your answer

Observation worksheet

Label both clockwise mesh currents and source polarities before writing KVL. Solve I1 and I2, then calculate the R3 branch current. Try equal sources and explain the zero shared current. E1=12 V; E2=6 V.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> Two clockwise mesh-current references. E1 is positive at the top; E2 is positive at the bottom. A clockwise walk therefore encounters a voltage rise in each source. Negative entered source values reverse the actual source voltage relative to that labelled polarity. R3 is shared; its downward reference current is I1−I2. This grounded planar network can also be solved by a one-node equation. Supports mesh-analysis preparation for Lab 2.

ECED / Module 1 · DC circuit fundamentals / LIVE LESSON

Source transformation

Compare voltage-source and current-source equivalents under the same load. Separate terminal equivalence from internal power behaviour.

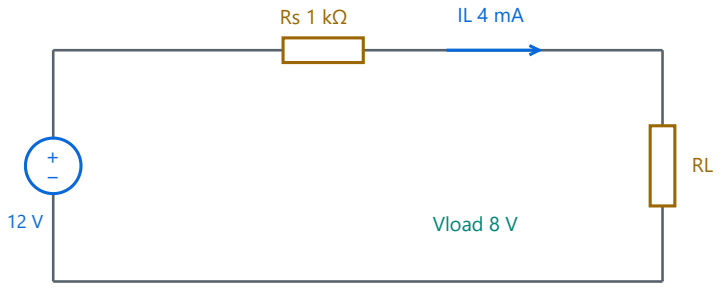
CO1 · Lab 2 support · Prerequisite: Ohm's law and KCL/KVL · Suggested activity: 15 minutes · Version 1.1 · Typeset mathematics

Norton source In	Load voltage	Load current
12 mA	**8 V**	**4 mA**
Vs / Rs	same in both circuits	same in both circuits

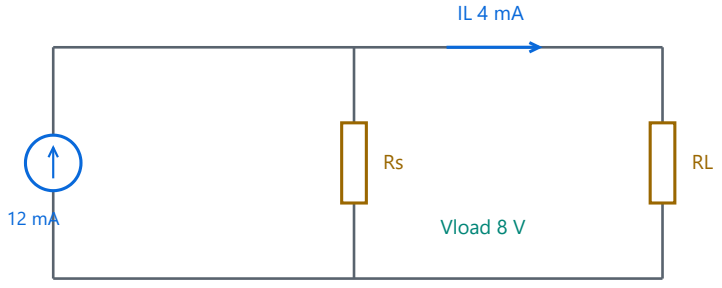
Different inside, equivalent at the terminals

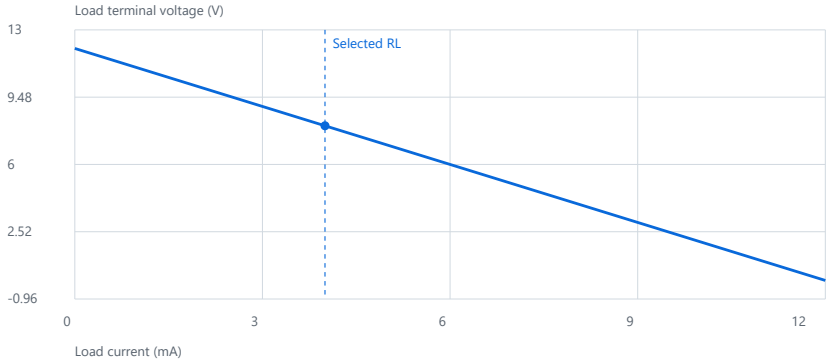
Keep the load outside the source-transformation boundary. Both source forms must present the same voltage and current to that load.

Voltage-source form — Rs is in series



Current-source form — Rs is in parallel





● Both equivalents share this terminal characteristic

Terminal equivalence does not mean identical internals. The voltage source carries 4 mA, while the current source supplies 12 mA. Their internal resistor losses are 16 mW and 64 mW, respectively.

Calculation step by step

RMS values for AC; SI units internally. Display rounding does not affect the calculations.

1 / TRANSFORM THE SOURCE

$$I_N = \frac{V_s}{R_s} = \frac{12}{1{,}000} = 12\,\text{mA}$$

$$R_N = R_s = 1\,\text{k}\Omega$$

2 / VOLTAGE-SOURCE FORM

$$I_L = \frac{V_s}{R_s + R_L} = 4\,\text{mA}$$

$$V_L = I_L R_L = 8\,\text{V}$$

3 / CURRENT-SOURCE FORM

$$I_L = \frac{I_N R_s}{R_s + R_L} = 4\,\text{mA}$$

$$I_{R_s} = \frac{V_L}{R_s} = 8\,\text{mA}$$

4 / TERMINAL CHARACTERISTIC

$$V_L = V_s - I_L R_s = I_N R_s - I_L R_s$$

Open-circuit VL = Vs = 12 V

Short-circuit IL = In = 12 mA

5 / KCL AND KVL CHECKS

$$I_N - I_{R_s} - I_L = 0\,\text{A}$$

$$V_s - I_L R_s - V_L = 0\,\text{V}$$

$$P_L = V_L I_L = 32\,\text{mW}$$

6 / INTERNAL POWER CHECK

Voltage form: source 48 mW = load + 16 mW

Current form: source 96 mW = load + 64 mW

Each circuit conserves power, even when its internal dissipation differs.

Check your understanding

Commit to a prediction, then check the explanation.

1. Where does Rs go when a series voltage source becomes a Norton source?

- ◯ It stays in series
- ◯ It becomes a shunt resistor across the current source

2. Which quantities must match for the same connected load?

- ◯ Load voltage and load current
- ◯ Every internal current and resistor loss

3. Numeric practice

Vs=12 V and Rs=1 kΩ are transformed to Norton form. A load RL=1 kΩ is attached. Find load voltage in volts.

Your answer

Observation worksheet

Record In, load voltage/current and both internal resistor losses for three load values. Explain why load behaviour matches even when internal losses do not. Vs=12 V; Rs=1000 Ω; RL=2000 Ω.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> An ideal voltage source Vs in series with positive Rs is terminal-equivalent to an ideal current source In=Vs/Rs in parallel with the same Rs. The same positive load RL is connected across the same output terminals. Only terminal voltage/current behaviour is preserved; the internal source currents and resistor losses need not match. The open-circuit and short-circuit intercepts are ideal limits, not extra zero-resistance control values. Supports source-transformation preparation for Module 1 and the network-theorem lab.

Superposition theorem

Separate two independent source contributions, add signed responses, and test why power cannot be superposed.

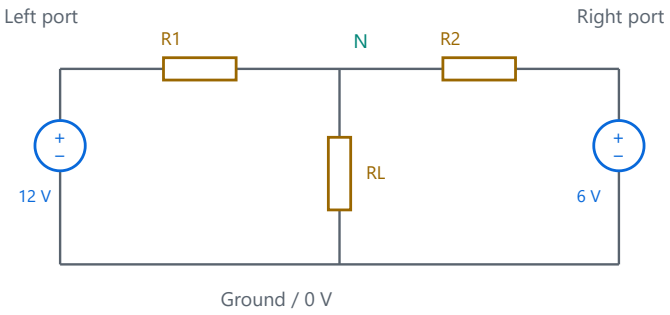
CO2 · Prerequisite: KCL, KVL and Ohm's law · Version 1.1 · Typeset mathematics

Load voltage	Load current	Load power
6 V	**6 mA**	**36 mW**
top relative to ground	positive toward ground	absorbed by load

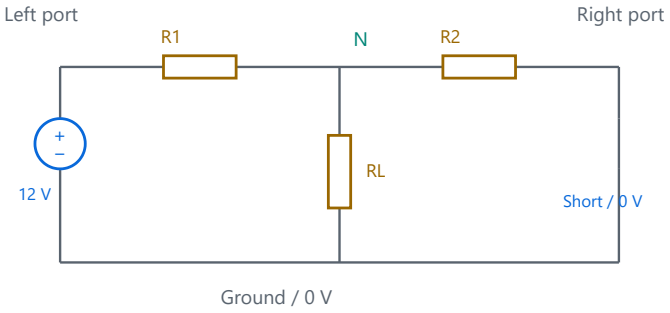
Superposition theorem

Read terminal labels and current reference directions before applying the theorem.

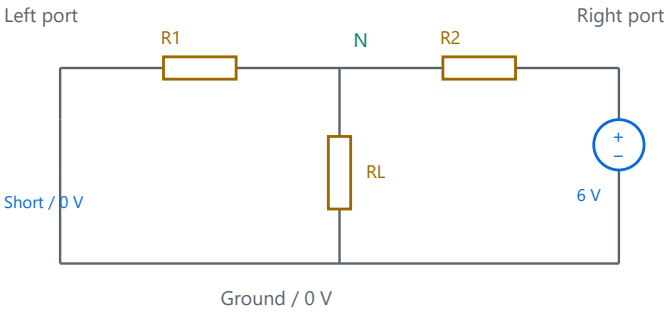
Both sources active (+ at the top; negative settings reverse polarity)

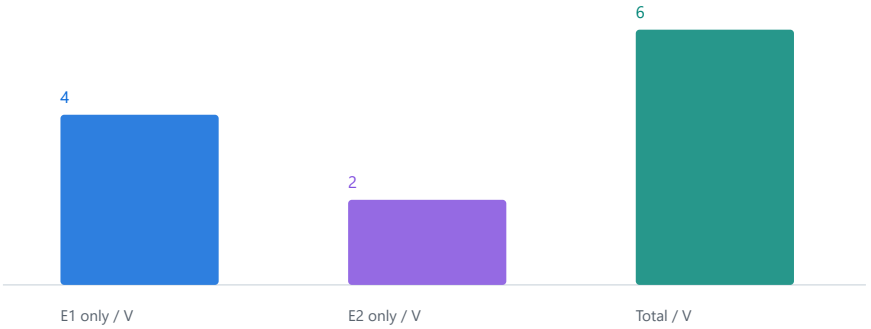


E1 alone: E2 becomes a short; R2 stays connected.



E2 alone: E1 becomes a short; R1 stays connected.





Change one control and compare the graph with the worked calculation.

Compare the signed voltage contributions. Keep the inactive source's series resistor in the circuit.

Calculation step by step

DC resistive circuit; SI units internally. Display rounding does not affect the calculations.

1 / CHOOSE RESPONSE

VL is measured from the top load terminal to ground; IL is downward through RL.

2 / KEEP E1 ACTIVE

Short ideal E2; retain R2. G = 1/R1 + 1/R2 + 1/RL.

$$V_{L1} = \frac{\frac{E_1}{R_1}}{G} = 4\,\mathrm{V}$$

3 / KEEP E2 ACTIVE

Short ideal E1; retain R1.

$$V_{L2} = \frac{\frac{E_2}{R_2}}{G} = 2\,\mathrm{V}$$

4 / ADD SIGNED RESPONSES

$$V_\mathrm{L} = V_{L1} + V_{L2} = 6\,\mathrm{V}$$

$$I_\mathrm{L} = \frac{V_\mathrm{L}}{R_\mathrm{L}} = 6\,\mathrm{mA}$$

5 / VERIFY DIRECT KCL

$$\frac{E_1 - V_\mathrm{L}}{R_1} + \frac{E_2 - V_\mathrm{L}}{R_2} - \frac{V_\mathrm{L}}{R_\mathrm{L}} = 0\,\mathrm{A}$$

6 / POWER IS NONLINEAR

$$P = \frac{V_\mathrm{L}^2}{R_\mathrm{L}} = 36\,\mathrm{mW}$$

$$P_1 + P_2 = 20\,\mathrm{mW}$$

The missing cross term is 2 VL_1 VL_2/RL. Power contributions do not generally add.

Check your understanding

Commit to a prediction, then check the explanation.

1. Can the powers of separate source responses generally be added?

- ◯ No; power is quadratic in voltage
- ◯ Yes; all electrical quantities superpose

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2. Does terminal equivalence require all internal currents to match?

- ◯ No; it concerns the defined terminal response
- ◯ Yes; every internal quantity must match

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3. Numeric practice

$E1=12$ V, $E2=6$ V and $R1=R2=RL=1$ kΩ. Find the total load voltage in volts.

Your answer

Observation worksheet

Record three parameter sets, predict each response, then compare calculation with the circuit. Include units and explain the theorem assumptions.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> Linear ideal resistive circuit. Both voltage sources have + at the top. Deactivate an independent voltage source with a short; deactivate an independent current source with an open. Retain resistors and dependent sources. This model contains no dependent sources. Module 2 / CO2; Lab 3 preparation.

ECED / Module 2 · Network theorems / LIVE LESSON

Thevenin and Norton equivalents

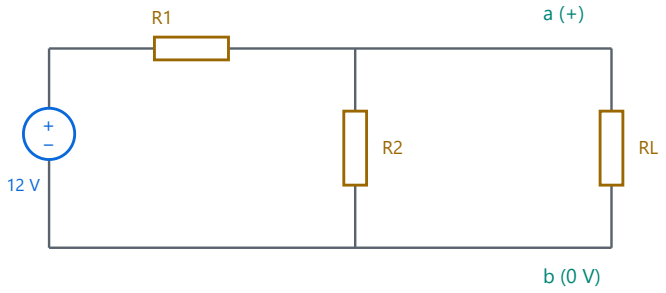
Remove the load, find the open-circuit voltage and equivalent resistance, then reconnect the load.

CO2 · Prerequisite: KCL, KVL and Ohm's law · Version 1.1 · Typeset mathematics

Load voltage	Load current	Load power
6 V	**3 mA**	**18 mW**
top relative to ground	positive toward ground	absorbed by load

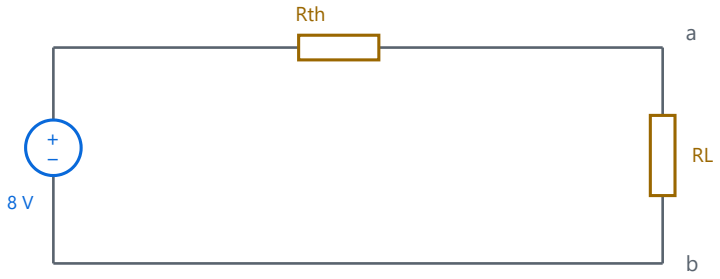
Thevenin and Norton equivalents

Read terminal labels and current reference directions before applying the theorem.

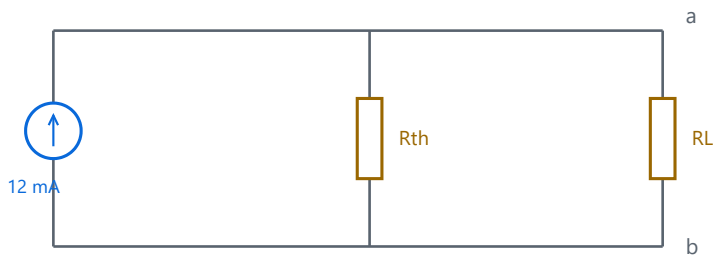


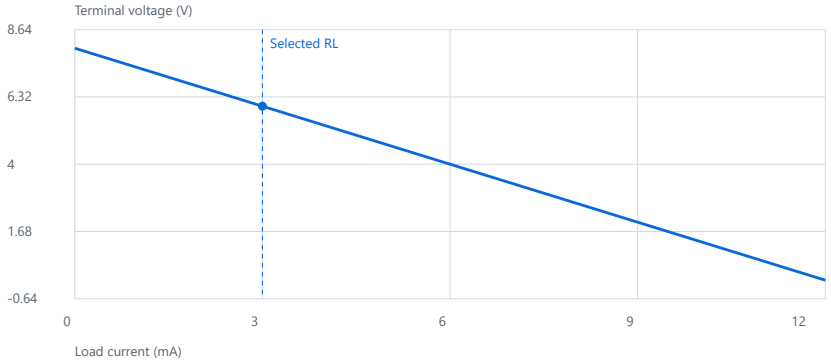
Equivalent seen at a–b: Vth in series with Rth, or In upward in parallel with Rth. RL connects across a–b.

Thevenin equivalent: connect the same RL at a–b.



Norton equivalent: source In supplies parallel Rth and RL.





Change one control and compare the graph with the worked calculation.

The two terminal equivalents predict the same loaded response as the original network.

Calculation step by step

DC resistive circuit; SI units internally. Display rounding does not affect the calculations.

1 / REMOVE THE LOAD

Open RL only. Keep R2 inside the source network.

2 / OPEN-CIRCUIT VOLTAGE

$$V_{\mathrm{th}} = \frac{V_{\mathrm{s}} R_2}{R_1 + R_2} = 8\,\mathrm{V}$$

3 / DEACTIVATE INDEPENDENT SOURCE

Short ideal Vs. Looking into the terminals, R1 and R2 are parallel.

$$R_{\mathrm{th}} = \frac{R_1 R_2}{R_1 + R_2} = 666.7\Omega$$

4 / NORTON CONVERSION

$$I_{\mathrm{N}} = \frac{V_{\mathrm{th}}}{R_{\mathrm{th}}} = 12\,\mathrm{mA}$$

Rn = Rth; In points upward into the output node.

5 / RECONNECT LOAD

$$I_{\mathrm{L}} = \frac{V_{\mathrm{th}}}{R_{\mathrm{th}} + R_{\mathrm{L}}} = 3\,\mathrm{mA}$$

$$V_{\mathrm{L}} = I_{\mathrm{L}} R_{\mathrm{L}} = 6\,\mathrm{V}$$

6 / CHECK ORIGINAL NETWORK

$$\frac{V_{\mathrm{s}} - V_{\mathrm{L}}}{R_1} - \frac{V_{\mathrm{L}}}{R_2} - I_{\mathrm{L}} = 0\,\mathrm{A}$$

$$P_{\mathrm{L}} = V_{\mathrm{L}} I_{\mathrm{L}} = 18\,\mathrm{mW}$$

Check your understanding

Commit to a prediction, then check the explanation.

1. When finding Rth here, what happens to the ideal voltage source?

- ◯ Replace it with a short circuit
- ◯ Replace it with an open circuit

---

2. Does terminal equivalence require all internal currents to match?

- ◯ No; it concerns the defined terminal response
- ◯ Yes; every internal quantity must match

---

3. Numeric practice

Vs=12 V, R1=1 kΩ, R2=2 kΩ, RL=2 kΩ. Find load voltage in volts.

Your answer

Observation worksheet

Record three parameter sets, predict each response, then compare calculation with the circuit. Include units and explain the theorem assumptions.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> Ideal independent voltage source and positive resistors. The output port is top node to ground; RL is external, R2 is internal. Dependent-source networks require a test-source method and are outside this fixed circuit. Module 2 / CO2; Lab 3 preparation.

ECED / Module 2 · Network theorems / LIVE LESSON

Reciprocity theorem

Move one voltage source between two branches and compare the current in the other branch with consistent directions.

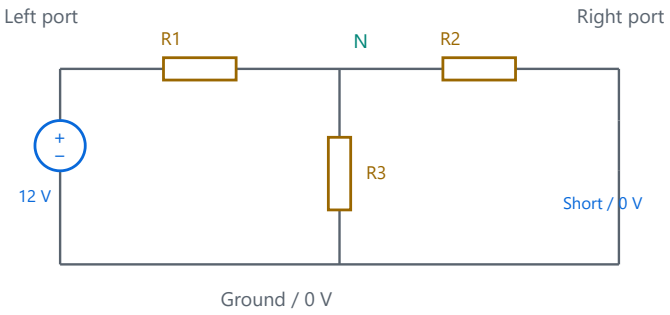
CO2 · Prerequisite: KCL, KVL and Ohm's law · Version 1.1 · Typeset mathematics

Forward current	Reverse current	Difference
3.273 mA	**3.273 mA**	**0 A**
right port response	left port response	reciprocity check

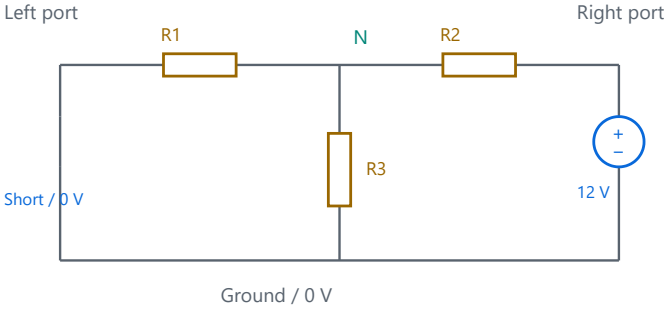
Reciprocity theorem

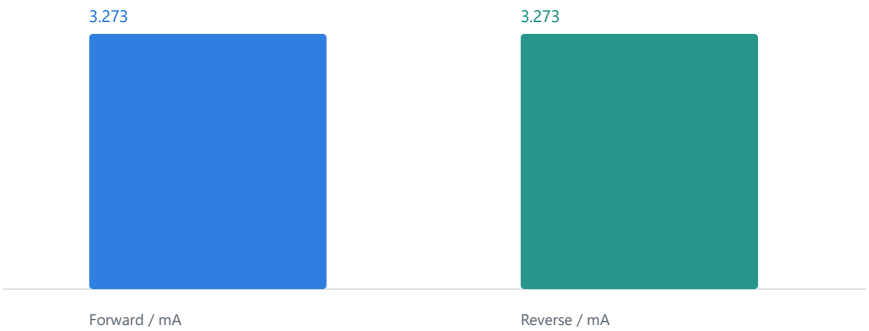
Read terminal labels and current reference directions before applying the theorem.

Forward: source at left, right source replaced with a short (0 V). Measure right current from N to the right port.



Reverse: source at right, left port shorted. Measure left current from N to the left port.





Change one control and compare the graph with the worked calculation.

The response currents match even though the internal node voltages can differ.

Calculation step by step

DC resistive circuit; SI units internally. Display rounding does not affect the calculations.

1 / FIX THE NETWORK

R1 joins left port to node N; R2 joins right port to N; R3 joins N to ground.

2 / EXCITE LEFT PORT

Apply E on the left (+ top). Right port is closed through an ideal ammeter.

$$V_{Na} = 6.545\,\mathrm{V}$$

3 / MEASURE RIGHT CURRENT

$$I_{\mathrm{right}} = \frac{V_{Na}}{R_2} = 3.273\,\mathrm{mA}$$

Reference direction: from N through R2 toward the shorted right port.

4 / SWAP SOURCE AND RESPONSE

Move the same source to the right. Short the left port through the ammeter.

$$V_{Nb} = 3.273\,\mathrm{V}$$

5 / MEASURE LEFT CURRENT

$$I_{\mathrm{left}} = \frac{V_{Nb}}{R_1} = 3.273\,\mathrm{mA}$$

Reference direction: from N through R1 toward the shorted left port.

6 / COMPARE TRANSFER

$$D = R_1R_2 + R_1R_3 + R_2R_3$$

$$\begin{aligned}\frac{I_{\mathrm{right}}}{E} &= \frac{I_{\mathrm{left}}}{E} \\ &= \frac{R_3}{D} \\ &= 272.7\,\mu\mathrm{S}\end{aligned}$$

$$\mathrm{Difference} \quad = 0\,\mathrm{A}$$

Check your understanding

Commit to a prediction, then check the explanation.

1. Must the two internal node voltages be equal after swapping ports?

- ◯ No; the transfer currents are equal under these conditions
- ◯ Yes; every internal quantity is unchanged

2. Does terminal equivalence require all internal currents to match?

- ◯ No; it concerns the defined terminal response
- ◯ Yes; every internal quantity must match

3. Numeric practice

E=12 V, R1=1 kΩ, R2=2 kΩ, R3=3 kΩ. Find the current at the opposite shorted port in mA.

Your answer

Observation worksheet

Record three parameter sets, predict each response, then compare calculation with the circuit. Include units and explain the theorem assumptions.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> Passive linear bilateral resistor network with one independent source at a time. The source and ammeter are interchanged at the specified ports with consistent polarities/directions. Internal voltages need not be equal. Reciprocity is not guaranteed for unilateral or active networks. Module 2 / CO2; supporting theorem activity.

ECED / Module 2 · Network theorems / LIVE LESSON

Maximum power transfer

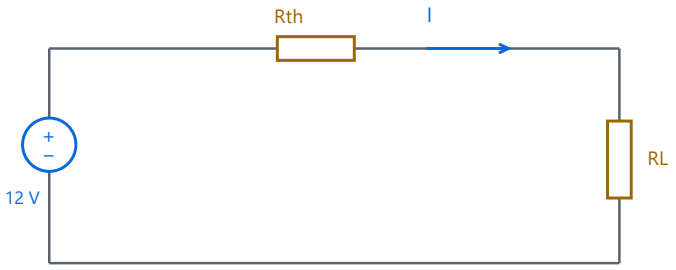
Sweep the load resistance and distinguish maximum load power from maximum efficiency.

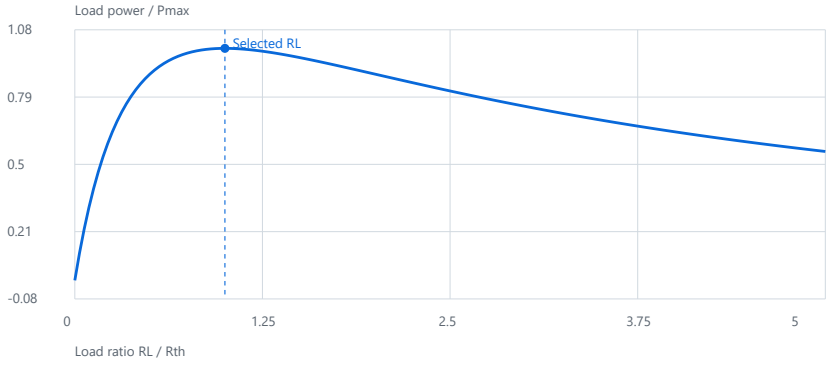
CO2 · Prerequisite: KCL, KVL and Ohm's law · Version 1.1 · Typeset mathematics

Load voltage	Load current	Load power
6 V	**6 mA**	**36 mW**
top relative to ground	positive toward ground	absorbed by load

Maximum power transfer

Read terminal labels and current reference directions before applying the theorem.





Change one control and compare the graph with the worked calculation.

The curve peaks at RL/Rth=1. Efficiency is 50% there, and rises for larger loads.

Calculation step by step

DC resistive circuit; SI units internally. Display rounding does not affect the calculations.

1 / WRITE LOAD CURRENT

$$I = \frac{V_{\mathrm{th}}}{R_{\mathrm{th}} + R_{\mathrm{L}}} = 6\,\mathrm{mA}$$

2 / WRITE LOAD POWER

$$\begin{aligned} P_{\mathrm{L}} &= I^2 R_{\mathrm{L}} \\ &= \frac{V_{\mathrm{th}}^2 R_{\mathrm{L}}}{(R_{\mathrm{th}} + R_{\mathrm{L}})^2} \\ &= 36\,\mathrm{mW} \end{aligned}$$

3 / DIFFERENTIATE

$$\frac{\mathrm{d}P_L}{\mathrm{d}R_L} = \frac{V_{\mathrm{th}}^2 (R_{\mathrm{th}} - R_L)}{(R_{\mathrm{th}} + R_L)^3}$$

The derivative changes from positive to negative at RL=Rth.

4 / MATCH RESISTANCES

$$\text{Optimum} \quad R_{\mathrm{L}} = R_{\mathrm{th}} = 1\,\mathrm{k\Omega}$$

$$P_{\max} = \frac{V_{\mathrm{th}}^2}{4R_{\mathrm{th}}} = 36\,\mathrm{mW}$$

5 / COMPARE EFFICIENCY

$$\begin{aligned} \eta &= \frac{P_{\mathrm{L}}}{P_{\mathrm{L}} + P_{\mathrm{internal}}} \\ &= \frac{R_{\mathrm{L}}}{R_{\mathrm{th}} + R_{\mathrm{L}}} \end{aligned}$$

Selected η = 50%; at the power maximum η=50%.

6 / CHECK ENERGY

$$P_{\mathrm{internal}} = I^2 R_{\mathrm{th}} = 36\,\mathrm{mW}$$

$$V_{\mathrm{th}} I - P_{\mathrm{L}} - P_{\mathrm{internal}} = 0\,\mathrm{W}$$

Larger RL raises efficiency but eventually makes load power approach zero.

Check your understanding

Commit to a prediction, then check the explanation.

1. What is efficiency at maximum load power in this resistive model?

- ◯ 50%
- ◯ 100%

---

2. Does terminal equivalence require all internal currents to match?

- ◯ No; it concerns the defined terminal response
- ◯ Yes; every internal quantity must match

3. Numeric practice

Vth=12 V and Rth=1 kΩ. Find maximum load power in mW.

Your answer

Observation worksheet

Record three parameter sets, predict each response, then compare calculation with the circuit. Include units and explain the theorem assumptions.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> DC ideal Thevenin source with positive internal and load resistances. Efficiency refers to the modeled source plus internal resistor. For AC complex impedances, maximum average power requires conjugate matching; this DC resistance model does not simulate that extension. Module 2 / CO2; supporting activity.

ECED / Module 2 · Network theorems / LIVE LESSON

Wye–Delta transformation

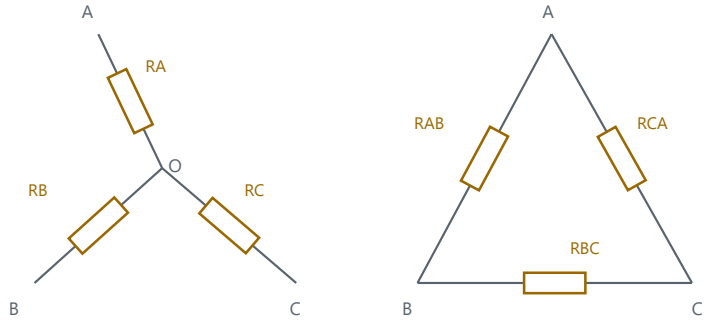
Convert a three-resistor star to a delta and verify all three terminal-pair resistance measurements.

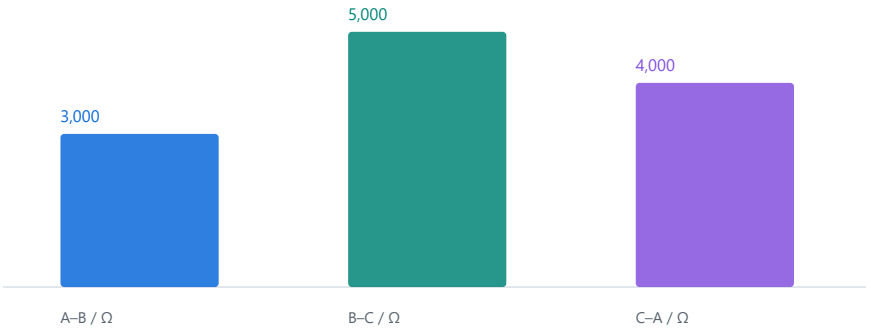
CO2 · Prerequisite: KCL, KVL and Ohm's law · Version 1.1 · Typeset mathematics

Delta RAB	Delta RBC	Delta RCA
3.667 kΩ	**11 kΩ**	**5.5 kΩ**
opposite RC	opposite RA	opposite RB

Wye–Delta transformation

Read terminal labels and current reference directions before applying the theorem.





Pair resistances are the same in both forms; see the calculations below.

For pair measurements, the third terminal is externally open. Its internal delta path still conducts.

Calculation step by step

DC resistive circuit; SI units internally. Display rounding does not affect the calculations.

1 / LABEL TERMINALS

RA joins A to centre O; RB joins B to O; RC joins C to O.

2 / FORM PAIR-PRODUCT SUM

$$S = R_\mathrm{A}R_\mathrm{B} + R_\mathrm{B}R_\mathrm{C} + R_\mathrm{C}R_\mathrm{A}$$
$$= 11{,}000{,}000\,\Omega^2$$

3 / STAR TO DELTA

$$R_{AB} = \frac{S}{R_\mathrm{C}}$$
$$= \frac{11000000\,\Omega^2}{3000\,\Omega}$$
$$= 3.667\,\mathrm{k\Omega}$$

$$R_{BC} = \frac{S}{R_\mathrm{A}}$$
$$= \frac{11000000\,\Omega^2}{1000\,\Omega}$$
$$= 11\,\mathrm{k\Omega}$$

$$R_{CA} = \frac{S}{R_\mathrm{B}}$$
$$= \frac{11000000\,\Omega^2}{2000\,\Omega}$$
$$= 5.5\,\mathrm{k\Omega}$$

4 / DELTA BACK TO STAR

$$T = R_{AB} + R_{BC} + R_{CA}$$

$$R_\mathrm{A} = \frac{R_{AB}R_{CA}}{T} = 1\,\mathrm{k\Omega}$$

$$R_\mathrm{B} = \frac{R_{AB}R_{BC}}{T} = 2\,\mathrm{k\Omega}$$

$$R_\mathrm{C} = \frac{R_{BC}R_{CA}}{T} = 3\,\mathrm{k\Omega}$$

5 / TEST A–B; C OPEN

Star: $R_\mathrm{A} + R_\mathrm{B} = 3\,\mathrm{k\Omega}$

Delta: $R_{AB} \parallel (R_{CA} + R_{BC}) = 3\,\mathrm{k\Omega}$

6 / TEST OTHER PAIRS

B–C: $5\,\mathrm{k\Omega} = 5\,\mathrm{k\Omega}$

C–A: $4\,\mathrm{k\Omega} = 4\,\mathrm{k\Omega}$

Terminal behaviour matches; individual internal resistor values and currents need not match.

Check your understanding

Commit to a prediction, then check the explanation.

1. Which star arm divides S when finding delta RAB?

- ◯ The opposite arm RC
- ◯ The arm RA

2. Does terminal equivalence require all internal currents to match?

- ◯ No; it concerns the defined terminal response
- ◯ Yes; every internal quantity must match

3. Numeric practice

A balanced star has RA=RB=RC=1 kΩ. Find each equivalent delta resistor in ohms.

Your answer

Observation worksheet

Record three parameter sets, predict each response, then compare calculation with the circuit. Include units and explain the theorem assumptions.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> Positive ideal resistors; fixed terminal labels A, B, C. Pair tests leave the third terminal externally open, while its internal delta path remains connected. Equivalence applies at all three external terminals. Module 2 / CO2; network reduction support.

ECED / Module 3 · AC circuit analysis / LIVE LESSON

Sinusoids, RMS and phase

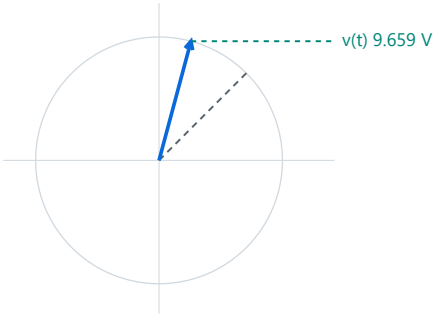
Connect a rotating peak vector with instantaneous voltage, waveform phase, frequency, period and RMS magnitude.

CO3 · Lab 6 preparation · Prerequisite: basic trigonometry · Suggested activity: 15 minutes · Version 1.1 · Typeset mathematics

RMS voltage	Period T	Voltage at cursor
7.071 V	**20 ms**	**9.659 V**
Vpk / √2	T = 1/f	at t = 0.125T

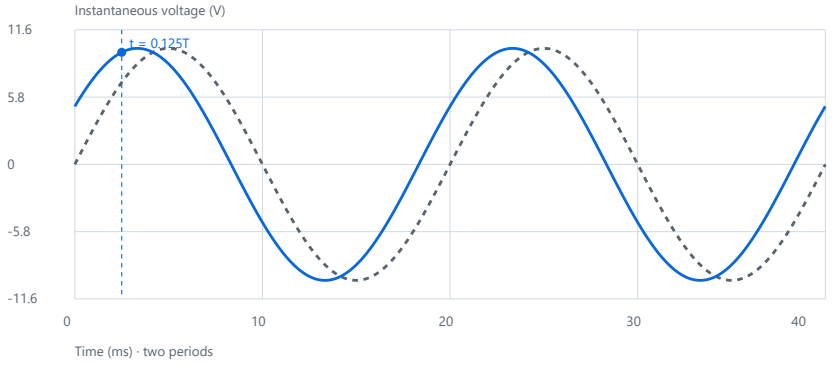
Connect the rotating arrow to the sine wave

The arrow is drawn at the selected time. Its vertical projection gives v(t). Its length represents peak voltage, not RMS voltage.



Peak arrow: 10 V · RMS magnitude: 7.071 V

φ = 30°; t = 2.5 ms



Selected phase · Zero-phase reference (dashed) · Instantaneous projection

Positive phase leads the reference. The selected wave reaches corresponding peaks earlier. Changing frequency changes the period, while RMS stays fixed if peak amplitude stays fixed.

Calculation step by step

RMS values for AC; SI units internally. Display rounding does not affect the calculations.

1 / FREQUENCY AND PERIOD

$$T = \frac{1}{f} = \frac{1}{50} = 20\,\text{ms}$$

$$\omega = 2\pi f = \frac{314.2\,\text{rad}}{s}$$

2 / PHASE CONVERSION

$$\varphi_{\text{rad}} = \frac{\varphi_{\text{deg}} \times \pi}{180} = 0.523599\,\text{rad}$$

$$v\,(t) = V_{\text{pk}} \sin\,(\omega t + \varphi)$$

3 / TIME CURSOR

$$t = 0.125T = 2.5\,\text{ms}$$

$$\omega t + \varphi = 1.309\,\text{rad}$$

$$v\,(t) = 10 \sin\,(1.309) = 9.659\,\text{V}$$

4 / RMS MAGNITUDE

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,\text{d}t}$$

$$\text{For a sine:}\quad V_{\text{rms}} = \frac{V_{\text{pk}}}{\sqrt{2}} = 7.071\,\text{V}$$

The rotating peak arrow must be divided by √2 to obtain the RMS circuit-phasor magnitude.

5 / OTHER WAVEFORM VALUES

$$\text{Vpeak-to-peak} \quad = 2V_{\text{pk}} = 20\,\text{V}$$

Average voltage over a complete period = 0 V

Average voltage is zero, but average squared voltage is not zero.

6 / CONSISTENCY CHECKS

$$fT = 1$$

$$V_{\text{rms}}^2 = \frac{V_{\text{pk}}^2}{2} = 50\,\text{V}^2$$

Increasing frequency leaves RMS unchanged when peak amplitude is held fixed.

Check your understanding

Commit to a prediction, then check the explanation.

1. Peak amplitude stays fixed while frequency doubles. Which statement is correct?

- ◯ Period halves; RMS voltage is unchanged
- ◯ Both period and RMS voltage double

2. The rotating peak arrow has length 10 V. What magnitude does the RMS circuit phasor use?

- ◯ 10 V
- ◯ 10/√2 V

3. Numeric practice

A pure sine wave has peak voltage 10 V. Find its RMS voltage in volts.

Your answer

Observation worksheet

Record peak, RMS, peak-to-peak value, period and cursor voltage. Double frequency and explain why RMS stays the same. Then apply +90° phase and compare with the dashed reference. Present source: 10 V peak, 50 Hz, phase 30°.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> Pure sinusoidal voltage $v(t)=V_{pk}\sin(\omega t+\varphi)$. Positive φ leads the zero-phase sine reference of the same frequency. The rotating arrow has PEAK magnitude so that its vertical projection equals instantaneous voltage; an RMS circuit phasor has magnitude $V_{pk}/\sqrt{2}$ instead. The grey trace and vector are zero-phase references. No DC offset, distortion, harmonics or transient is included. RMS=peak/√2 is specific to a sine wave, not an arbitrary waveform. Supports Module 3 and Lab 6 preparation.

CIRCUIT NOTES / INTERACTIVE EXPLORER

Phasor Lab

Change the circuit. See voltage and current vectors. Understand the phase.

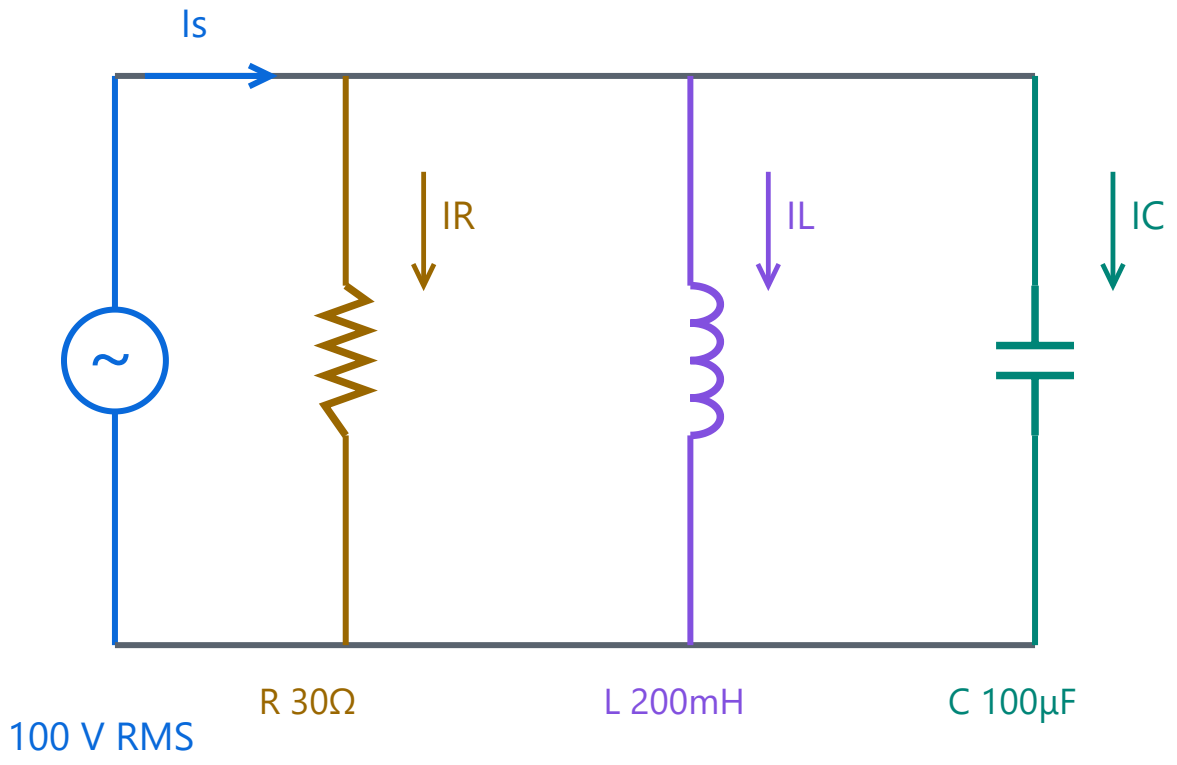
Series + parallel · RMS

RMS current	Current phase θ	Average power
3.676 A	**+24.94°**	**333.3 W**
|Z| = 27.20 Ω	Current leads source	Power factor 0.907

The parallel circuit

RLC parallel

Each branch has the same voltage. Source current splits between the branches.

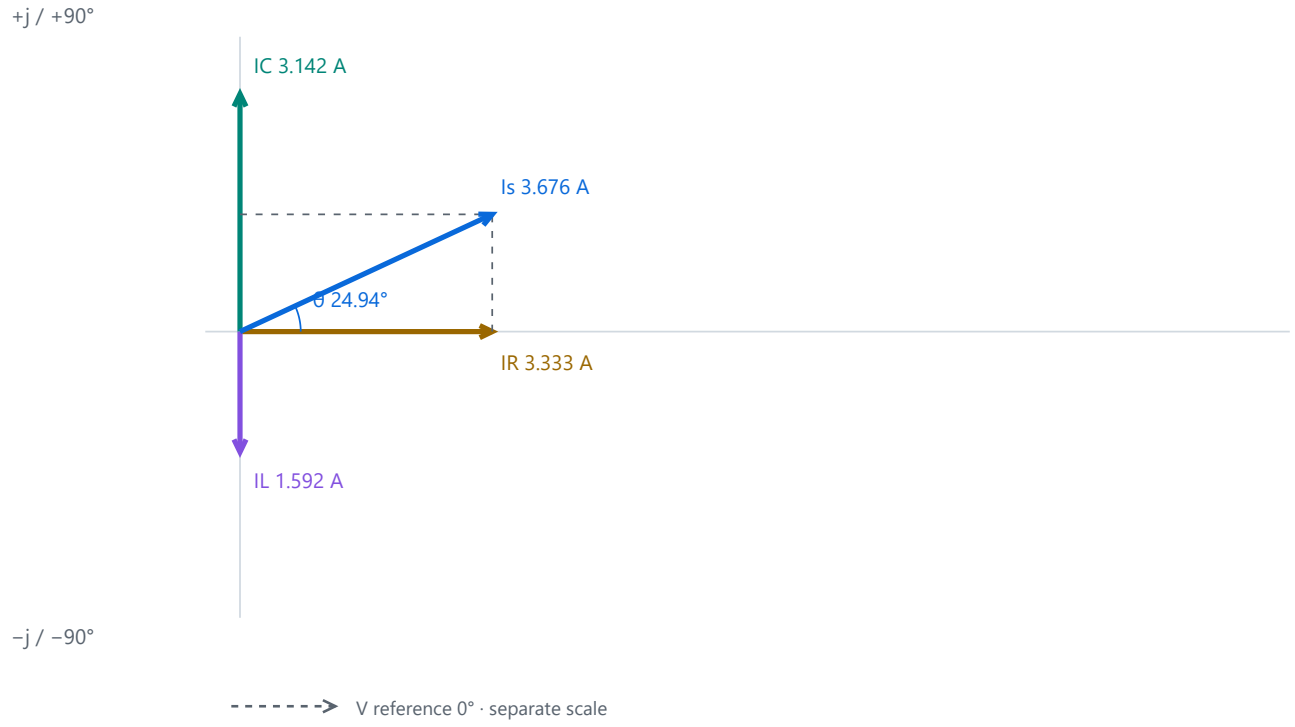


Current phasors

100 px = 2.293 A

Voltage is the 0° reference. The blue arrow is total source current.

IR IL IC Is



Capacitive: current leads by 24.94°. IC exceeds IL, so the source current points upward.

Auto-scaled on each change. Current arrows share a scale. The dashed voltage arrow is a separate reference.

Your calculation, step by step

Every value below updates with the controls. Calculations retain full precision internally.

1 / SHARED VOLTAGE

All connected branches have the same RMS voltage.

$$V_{\mathrm{R}} = V_{\mathrm{L}} = V_{\mathrm{C}} = V_{\mathrm{s}} = 100.00\,\mathrm{V}$$

Use KCL: Is = IR + IL + IC (phasor sum).

2 / REACTANCE AND ADMITTANCE

$$X_{\mathrm{L}} = \omega L = 62.83\Omega$$

$$X_{\mathrm{C}} = \frac{1}{\omega C} = 31.83\Omega$$

$$Y = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

Omit 1/ωL when the inductor branch is absent.

3 / BRANCH CURRENTS

$$I_{\mathrm{R}} = \frac{V}{R} = \frac{100.00}{30.00} = 3.333\,\mathrm{A}{\rightarrow}$$

$$I_{\mathrm{L}} = \frac{V}{X_{\mathrm{L}}} = 1.592\,\mathrm{A}{\downarrow}$$

$$I_{\mathrm{C}} = \frac{V}{X_{\mathrm{C}}} = 3.142\,\mathrm{A}{\uparrow}$$

Omitted branches carry zero current.

4 / ADD CURRENT PHASORS

$$I_{\mathrm{s}} = I_{\mathrm{R}} + j\,(I_{\mathrm{C}} - I_{\mathrm{L}})$$

$$I_{\mathrm{s}} = 3.333 + j1.550\,\mathrm{A}$$

$$|I_{\mathrm{s}}| = \sqrt{I_{\mathrm{R}}^2 + (I_{\mathrm{C}} - I_{\mathrm{L}})^2} = 3.676\,\mathrm{A}$$

5 / CURRENT PHASE

$$\theta = \mathrm{atan2}\,(I_{\mathrm{C}} - I_{\mathrm{L}}, I_{\mathrm{R}})$$

$$\theta = 24.94^{\circ}$$

Positive θ means current leads voltage. Negative θ means current lags. θ = −φ, where φ is the impedance angle.

6 / POWER AND EQUIVALENT IMPEDANCE

$$P = \frac{V^2}{R} = \frac{100.00^2}{30.00} = 333.3\,\mathrm{W}$$

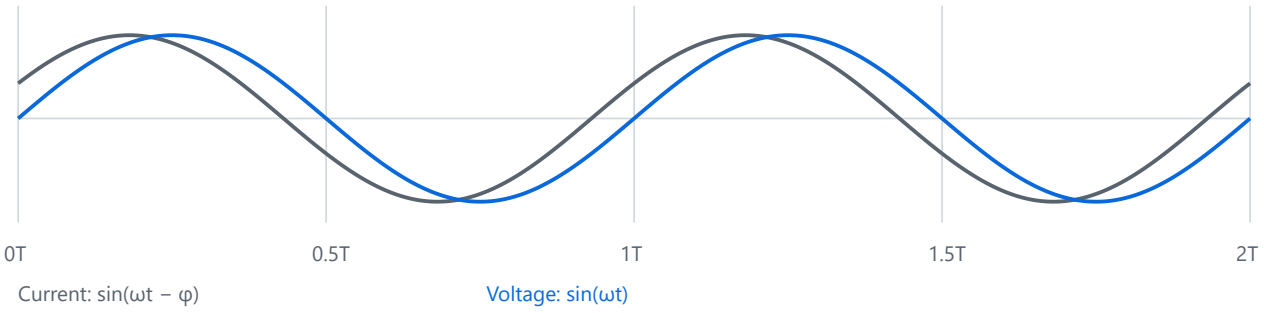
$$|Z_{\mathrm{eq}}| = \frac{V}{|I_{\mathrm{s}}|} = 27.20\Omega$$

Power factor = IR/|Is| = 0.907

At ideal parallel resonance, IL = IC; source current is minimum at V/R.

What does the phase look like in time?

Two cycles. Voltage and current are normalized separately to compare timing, not magnitude.



ECED / Module 3 · AC circuit analysis / LIVE LESSON

Complex power and power factor

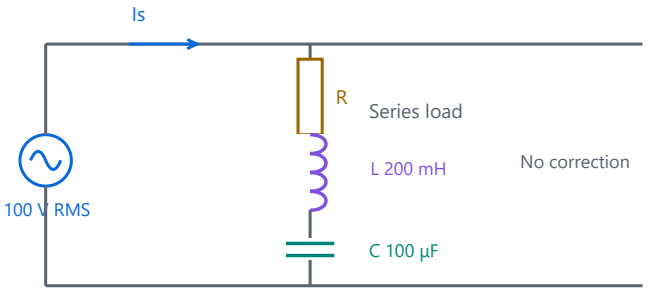
Connect the power triangle to source current and instantaneous power. Explore ideal shunt-capacitor correction and overcorrection.

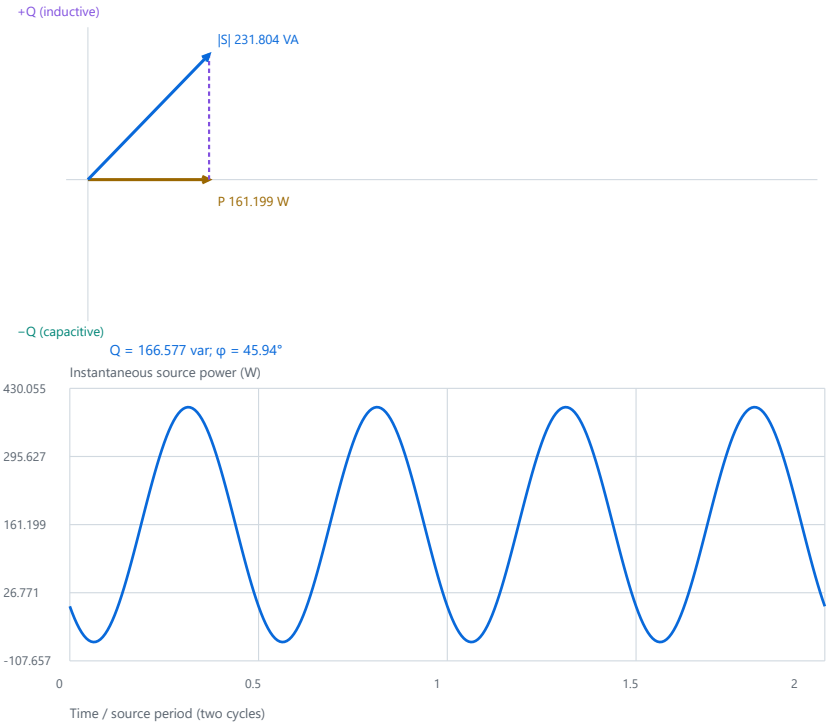
CO3 · Lab 6 extension · Prerequisite: RMS values and phasors · Suggested activity: 20 minutes

Useful real power P	Source power factor	RMS source current
161.199 W	**0.6954**	**2.318 A**
unchanged by ideal correction	lagging	before: 2.318 A

Follow the power triangle

The load is series RLC. The correction capacitor is a separate shunt branch. Positive Q points upward; negative Q points downward.





● P: real power ● Q: reactive power ● |S|: apparent power / p(t)

The source sees an inductive load. Current lags voltage. A suitable shunt capacitor can reduce source reactive power.

Calculation step by step

RMS values for AC; SI units internally. Display rounding does not affect the calculations.

1 / LOAD IMPEDANCE

$$X = \omega L - \frac{1}{\omega C} = 31.001\,\Omega$$

Omit the capacitor reactance term if series C is zero.

$$Z_{\text{load}} = 30 + j31.001\,\Omega$$

2 / COMPLEX LOAD POWER

$$S_{\text{load}} = V I_{\text{load}}^{*} = P + jQ_{\text{load}}$$

$$P = \frac{V^2 R}{R^2 + X^2} = 161.199\,\text{W}$$

$$Q_{\text{load}} = \frac{V^2 X}{R^2 + X^2} = 166.577\,\text{var}$$

3 / SHUNT CAPACITOR

$$Q_{\text{c}} = -\omega C_{\text{comp}} V^2 = 0\,\text{var}$$

$$Q_{\text{source}} = Q_{\text{load}} + Q_{\text{c}} = 166.577\,\text{var}$$

The ideal source holds load voltage fixed, so load P does not change.

4 / SOURCE APPARENT POWER

$$|S_{\text{source}}| = \sqrt{P^2 + Q_{\text{source}}^2} = 231.804\,\text{VA}$$

$$I_{\text{s}} = \frac{|S_{\text{source}}|}{V} = 2.318\,\text{A}$$

5 / PHASE AND POWER FACTOR

$$\varphi = \text{atan2}\,(Q_{\text{source}}, P) = 45.94^{\circ}$$

$$\text{PF} = \frac{P}{|S_{\text{source}}|} = 0.69541$$

Positive φ: lagging current. Negative φ: leading current.

6 / UNITY-PF CORRECTION

$$C_{\text{comp}} = \frac{Q_{\text{load}}}{\omega V^2}, \quad Q_{\text{load}} > 0$$

$$C_{\text{comp}} = 53.023092\,\mu\text{F}$$

Too much correction can increase source current again.

Check your understanding

Commit to a prediction, then check the explanation.

1. An ideal shunt capacitor corrects an inductive load. Which stays unchanged?

- ◯ Original load real power at fixed voltage
- ◯ Source reactive power
- ◯ Source RMS current

2. A capacitor is added to a load that already has leading power factor. What happens?

- ◯ It always brings PF to unity
- ◯ It makes net Q more negative; correction may worsen

3. A worked numeric check

A sinusoidal load consumes P=800 W and Q=600 var. Find apparent power in VA.

Your answer

Observation worksheet

Start with no correction capacitor. Record P, Q, |S|, PF and source current. Calculate unity-PF capacitance if the load is inductive. Apply it, then overcorrect and explain why current rises again. Compare phasor measurements with Lab 6.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> Sinusoidal steady state with RMS values. The load is an ideal series RLC network; zero L or C omits that series component (a wire). The correction capacitor is a separate ideal shunt branch across the supply. P is absorbed real power; Q>0 is inductive and Q<0 capacitive. Ideal shunt correction changes source Q and current, not load voltage or P. No capacitor loss, harmonics or switching transients are modelled. Supports Module 3 and interpretation of Lab 6.

RC and RL transients

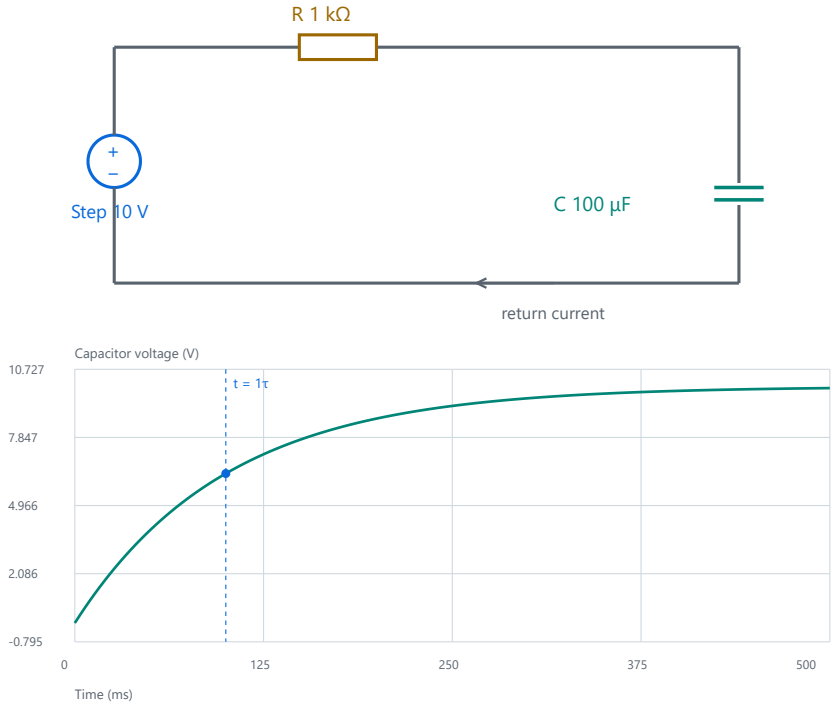
Explore source-step responses with initial conditions. Measure time constants and inspect voltage, current and stored energy.

CO4 · Labs 4 and 7 · Prerequisite: KCL/KVL and exponential functions · Suggested activity: 20 minutes

Time constant τ	State at cursor	Final state
100 ms	**6.321 V**	**10 V**
τ = RC	at t = 100 ms	limit as t → ∞

Capacitor voltage after a source step

The stored state cannot jump at t = 0. The graph spans five time constants; use the time cursor to inspect the response.



State response Time cursor

At one τ, 63.2% of the total change is complete. At five τ, 99.33% is complete. This applies to decay as well as charging; it does not always mean 63.2% of the source voltage.

Calculation step by step

RMS values for AC; SI units internally. Display rounding does not affect the calculations.

1 / INITIAL AND FINAL VALUES

$v_{\mathrm{C}}\left(0+\right) = 0\,\mathrm{V}$

$v_{\mathrm{C}}\left(\infty\right) = V_{\mathrm{s}} = 10\,\mathrm{V}$

2 / TIME CONSTANT

$\tau = R \times C = 1{,}000 \times 0.0001000 = 100\,\mathrm{ms}$

3 / GENERAL RESPONSE

$x\left(t\right) = x_{\mathrm{f}} + \left(x_{0} - x_{\mathrm{f}}\right) e^{-\frac{t}{\tau}}$

$v_{\mathrm{C}}\left(t\right) = 10 + \left(0 - 10\right) e^{-\frac{t}{\tau}}$

Use SI units in this substitution: V for RC, A for RL.

4 / AT THE TIME CURSOR

$t = 1\tau = 100\,\mathrm{ms}$

$v_{\mathrm{C}}\left(t\right) = 6.321\,\mathrm{V}$

Circuit current = 3.679 mA

5 / KVL AT THE CURSOR

$V_{\mathrm{s}} = v_{\mathrm{R}} + v_{\mathrm{C}}$

$10 = 3.678794 + 6.321206\,\mathrm{V}$

6 / STORED ENERGY

$E_{\mathrm{C}} = \frac{1}{2} C v_{\mathrm{C}}^{2} = 1.998\,\mathrm{mJ}$

The storage-element energy can decrease during discharge; the resistor dissipates energy.

Check your understanding

Commit to a prediction, then check the explanation.

1. What is continuous at an ideal source step in this circuit?

- ◯ Capacitor current and inductor voltage
- ◯ Capacitor voltage and inductor current

---

2. What happens at one time constant?

- ◯ The response reaches exactly its final value
- ◯ 63.2% of the total state change is complete
- ◯ The state must equal 63.2% of the source voltage

3. A worked numeric check

A capacitor has C=100 μF and series R=1 kΩ. Find τ in milliseconds.

Your answer

Observation worksheet

Use the RC preset. Predict x(0+), x(∞) and τ. Record the state at 0τ, 1τ, 3τ and 5τ. Compare the time constant with LTspice (Lab 4) or measurement (Lab 7).

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> Ideal source step at t=0 through series R. RC state is capacitor voltage; RL state is inductor current. Initial stored energy is allowed. τ=RC or L/R for this fixed single-resistor topology; more general circuits require the resistance seen by the storage element. Supports Labs 4 and 7. The plot shows 0–5τ, not a finite-time claim of exact settling.

ECED / Module 4 · Transients and frequency response / LIVE LESSON

Resonance, bandwidth and Q

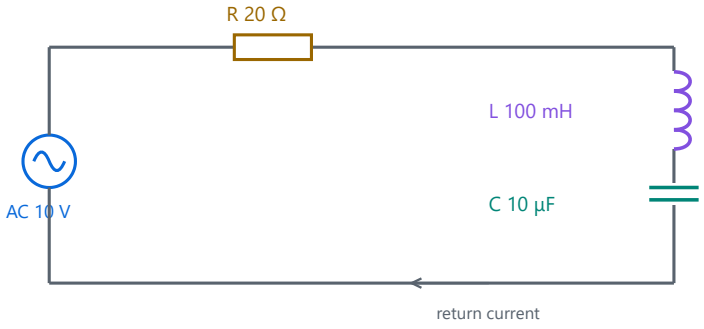
Compare series current resonance with parallel impedance resonance. Explore exact bandwidth boundaries and quality factor.

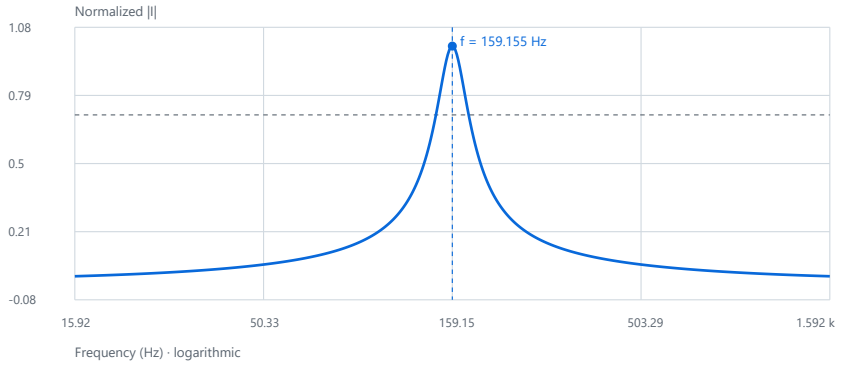
CO4 · Labs 5 and 8 · Prerequisite: Phasor Lab · Suggested activity: 20 minutes

Resonance f0	Bandwidth	Quality factor Q
159.155 Hz	**31.831 Hz**	**5**
XL = XC	f2 − f1	f0 / bandwidth

Series current resonance

Curve = |I| / Imax, fixed-voltage drive. Dashed line is 1/√2 of the peak current.





● Normalized current ● 1/√2 threshold

At resonance, current is maximum: 500 mA. At the selected frequency (159.155 Hz), current is 500 mA. Increasing series R widens the bandwidth and lowers Q.

Calculation step by step

RMS values for AC; SI units internally. Display rounding does not affect the calculations.

1 / DEFINE THE RESPONSE

Fixed RMS voltage source → series current magnitude. Half-current-squared points are resistor half-power points.

2 / RESONANCE

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$f_0 = 159.155\,\text{Hz}$$

$$X_\text{L} = X_\text{C} at f_0$$

3 / BANDWIDTH AND Q

$$\text{BW} = \frac{R}{2\pi L} = 31.831\,\text{Hz}$$

$$Q = \frac{f_0}{\text{BW}} = 5$$

4 / RESPONSE BOUNDARIES

$$\Delta\omega = 2\pi\text{BW}; \quad \omega_0 = 2\pi f_0$$

$$\omega_{1,2} = \frac{\sqrt{(\Delta\omega)^2 + 4\omega_0^2} \mp \Delta\omega}{2}$$

$$f_1 = 144.033\,\text{Hz}; \quad f_2 = 175.864\,\text{Hz}$$

These exact roots need not be symmetric around f0 on a linear frequency axis.

5 / AT SELECTED FREQUENCY

$$f = 1 \times f_0 = 159.155\,\text{Hz}$$

$$|Z| = 20\Omega$$

$$|I_\text{s}| = \frac{V_\text{s}}{|Z|} = 500\,\text{mA}$$

6 / CHECK THE BANDWIDTH

$$f_2 - f_1 = 31.831\,\text{Hz}$$

Response at f1 and f2 = 0.7071 of peak

$$20\log_{10}\left(\frac{1}{\sqrt{2}}\right) = -3.0103\,\text{dB}$$

Check your understanding

Commit to a prediction, then check the explanation.

1. In ideal series RLC with fixed voltage, current at resonance is...

- ◯ maximum
- ◯ minimum
- ◯ always zero

2. For ideal shunt RLC under fixed voltage, resistor power during a frequency sweep is...

- ◯ constant at V^2/R
- ◯ half its peak at f1 and f2
- ◯ zero at resonance

3. A worked numeric check

Find f0 in Hz for L=100 mH and C=10 µF. R does not affect ideal f0.

Your answer

Observation worksheet

Use series RLC. Record f1=144.033 Hz, f0=159.155 Hz, f2=175.864 Hz and Q=5. State the plotted response before interpreting the −3 dB points. Compare with Lab 5 or 8.

My prediction and reason

One measured or simulated result, with units

What changed, and why?

Model and lab connection

> Ideal series RLC, or three independent R/L/C branches in parallel (R is a shunt resistance, not coil resistance). Series curve: |I|/Imax for a fixed-voltage drive; resistor power is half at the −3.01 dB points. Parallel curve: |Z|/R, equivalent to normalized output voltage under fixed-current drive. With the fixed-voltage source shown here, parallel resistor power is constant and source current is minimum at resonance. Do not confuse these different response definitions. Supports Labs 5 and 8.